



2019-2023 VOLCANIC TEMPERATURE MONITORING USING GROUND THERMOMETERS & DRONE THERMAL SCANNING: AN APPLICATION ON HELLENIC VOLCANOES

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Abstract:

The Nisyros volcano located on the southeast part of the Aegean Plate boundary belongs to the Hellenic Volcanic Arc that results from the culmination of magmatic activity by the African Plate subducting beneath the Aegean overriding micro-plate. Hellenic Volcanoes are attractive touristic locations for millions of tourists annually. Volcanic monitoring techniques may promote a sustainable touristic environment, especially with the improved technological advantages of thermal drone scanning. A series of repeated survey flights and annual thermal mappings of an active crater, such as in Nisyros, is presented in this research. Studying changes in the craters' surface, surrounding landscape, and temperature changes is necessary to analyze the amount of erosion, stability, and volcanic activity. Drone imaging and the use of photogrammetry mapping software allows us to reconstruct a digital elevation model of a volcano; including but not limited to: Methana, Sousaki, Santorini, Milos, and Nisyros on a thermal and visual spectrum. Afterwards, the reconstruction can be compared and subtracted from previous years of digital elevation models and thermal footprint data. The goal is to detect any thermal and land changes due to volcanic or erosion activity based on the previous models.

A series of repeated flights using optical and thermal imaging cameras at the active Hellenic Volcanoes has provided a valuable database of the thermal signature from each volcano. In this research, we present the active volcanic crater of Stefanos on Nisyros Island. Using a DJI Mavic 3T with a thermal camera uncooled VOx Microbolometer (40 mm and 640x512px resolution), we mapped high-resolution surface temperatures with less than 5 cm per pixel. Furthermore, we observe no significant spatial changes in the surface temperature between 2019 and 2023. This methodology can also be applied to other volcanoes with annual periodic scanning to investigate normal volcanic activity.

Background Information:

- The Hellenic Volcanic Arc (HVA) contains the active volcanic islands located in the southern Aegean Sea (Greece) including but not limited to Sousaki, Santorini, Milos, and Nisyros.
- The HVA arc was formed around 16 million years ago due to the subduction of the African Plate beneath the Aegean Sea Plate, and it is one of the most seismically active areas in the world with the latest eruptions occurred in 1888 at Nisyros and 1950 at Santorini.



Figure 1: Volcanic eruption in Santorini (1923).

- Due to rapid changes in volcanic and seismic activity in this region, the volcanic islands located on the southern margin of the Aegean Sea Plate require annual evaluations to detect changes on volcanic activity.
- Monitoring of the active volcanic islands is critical to the European air traffic, touristic activity, and infrastructure because volcanic activity may pose a potential hazard to the people and the future habitability.
- Students and faculty members from Hofstra University in New York with collaboration of the National Kapodistrian University of Athens in Greece have applied multiple methods of volcanic monitoring that include annual drone mapping on the thermal and visual spectrum.

Method/Materials:

- Two different types of drones and cameras were used to take images of the Stefanos crater in Nisyros. The first drone was used to take pictures of the crater on the visual spectrum (DJI MAVIC PRO 2) while the second drone (DJI MAVIC 3T) was used to take pictures on the thermal spectrum with a thermal camera.
- Visual images of the surface of the Stefanos crater were taken in the middle of the day whereas thermal images of the crater were taken at night when heat from the sun is most likely to not affect the clarity of the scans.
- Hundreds of these images were then processed using Global Mapper and Pix4D, a photogrammetry software for professional drone mapping to create a digital elevation (DEM) and thermal surface (TSM) models of the crater.
- DEM & TSM models were then compared to the models of previous years to see if any significant changes have occurred at the crater.



Figures 2, 3, & 4: (left) Hofstra University undergraduate students Dylan Hubbs and Jett Hope driving the thermal drone at night at the Stefanos Crater in Nisyros, Greece; (middle) Dr. Kyriakopoulos (National Kapodistrian University of Athens) and Eleni Tsiouris (Townsent Harris High School student, NY) at Nisyros crater extracting volcanic gas samples; (right) Dr. Marsellos (Hofstra University), Dr. Kyriakopoulos, Eleni Tsiouris, Jett Hope and Dylan Hubbs setting up new thermometer sensors at the Stefanos crater.

Results:

- Thermal reconstructions after subtraction from previous years indicate that the crater seemed to be warmer in 2019 but with few exceptional higher temperatures in 2022, while in 2023 a higher resolution shows similar spatial results.
- There is a difference of the maximum temperature by 2 °C.
- Digital Elevation Models are still in processing to compare the elevation from current and previous years to reveal changes at the crater and surrounding areas.

Discussion/Conclusion:

- Based on the results from the thermal scanning of Nisyros Stefanos crater, we can conclude that other than a slight difference in temperature, most of the heat emissions at the Stefanos crater thermal map seem to be similar.
- The temperature map shows an increase of the maximum temperature by 2°C slightly but it is possible that it may be accounted an atmospheric influence that requires further investigation.
- With the technological use of mapping software such as Global Mapper and Pix4D, the ability to create DEM and DTM models using multiple drones as a medium helps to create a visual environment of trackable changes occurring along the Stefanos Crater. Evaluations and future assessments can be performed annually to monitor any changes of the crater in order to provide a useful analysis and keep the inhabitants of the island safe.

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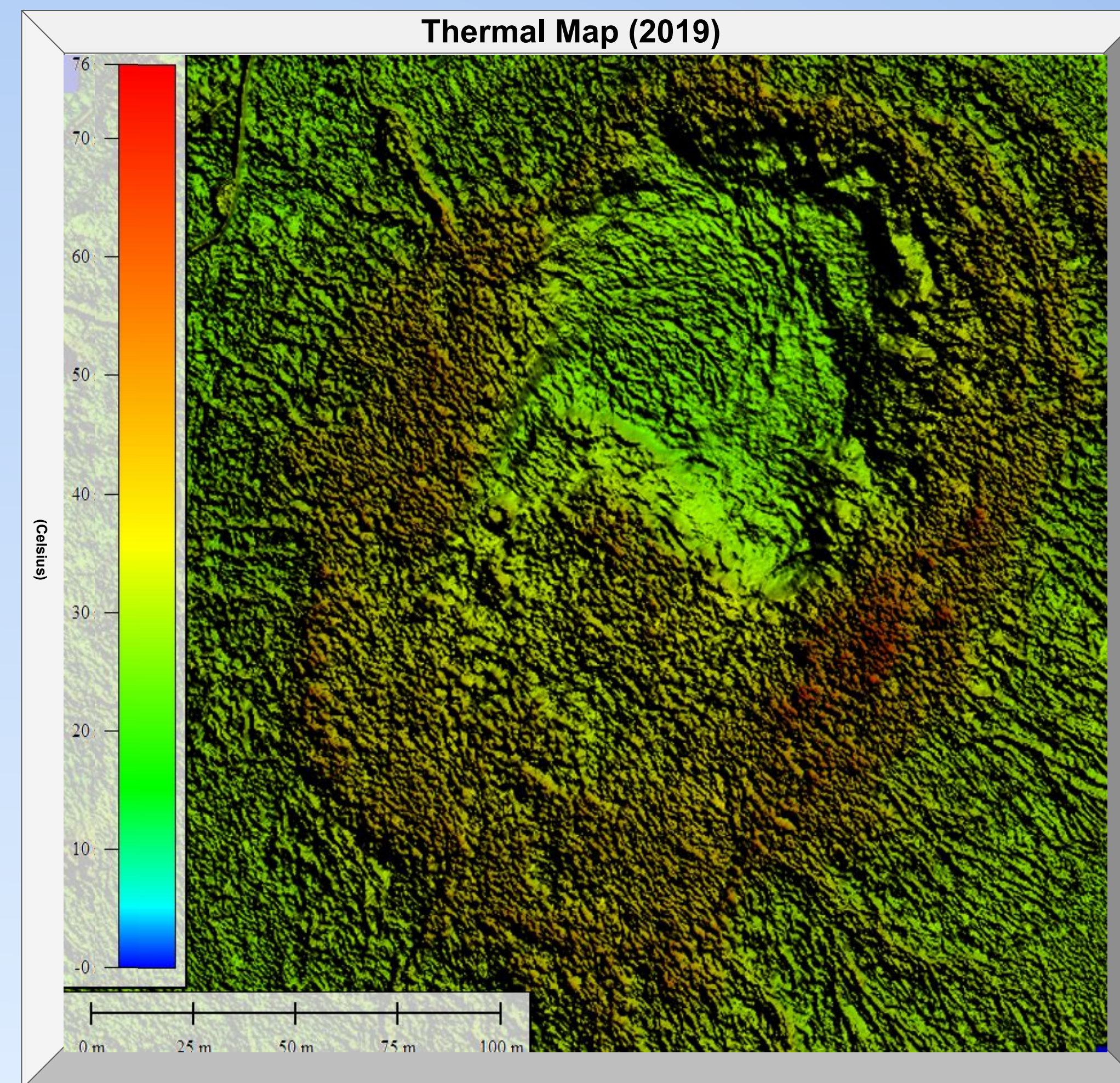


Figure 5: Thermal map made in Pix4D and reclassified in Global Mapper v.23.1 with a vertical scale showing the maximum and minimum temperatures scanned on Stefanos crater at Nisyros in 2019.

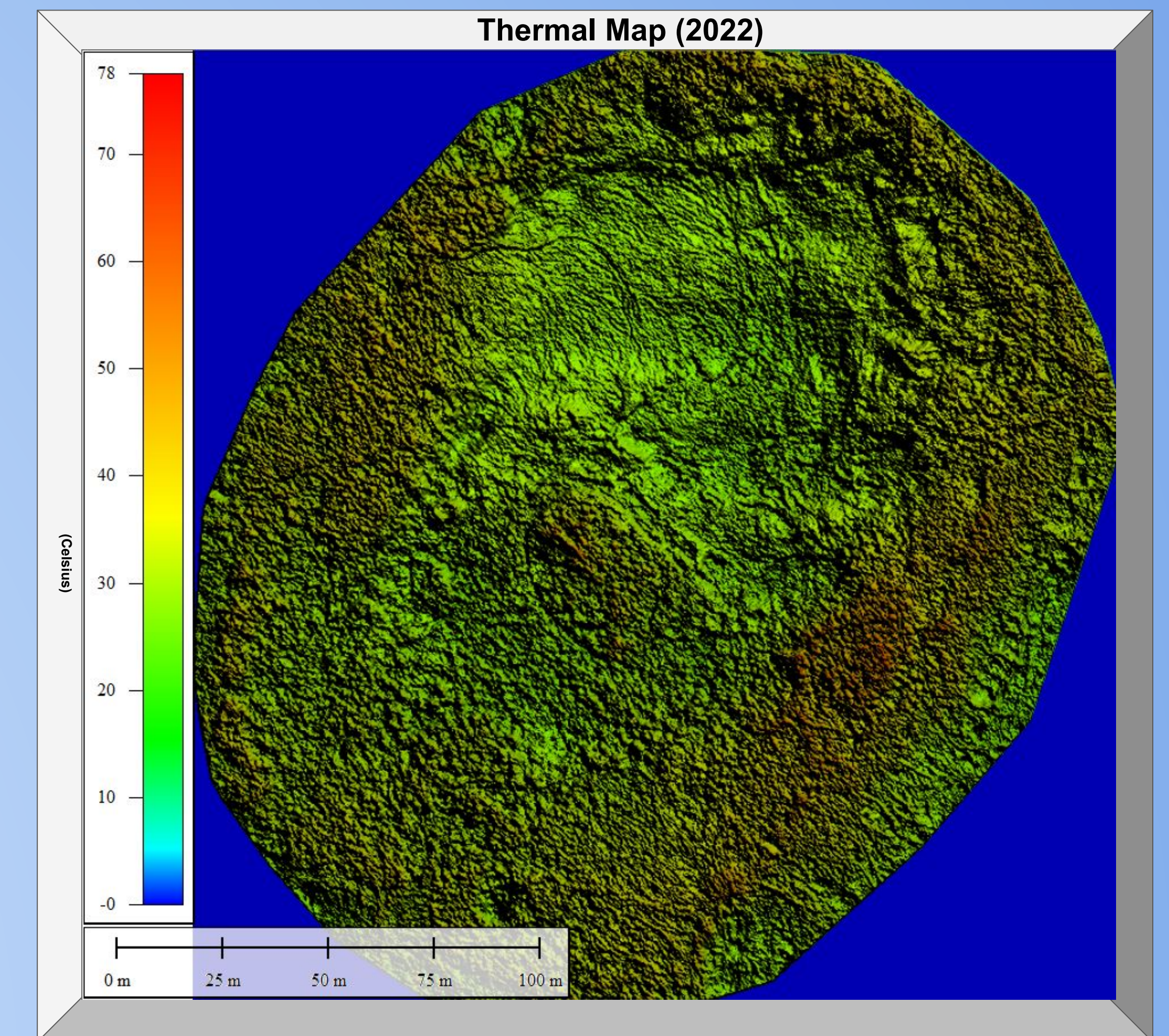


Figure 6: Thermal map made within the same parameters, equipment and methodology by scanned thermal images in 2022 (DJI Matrice M100).

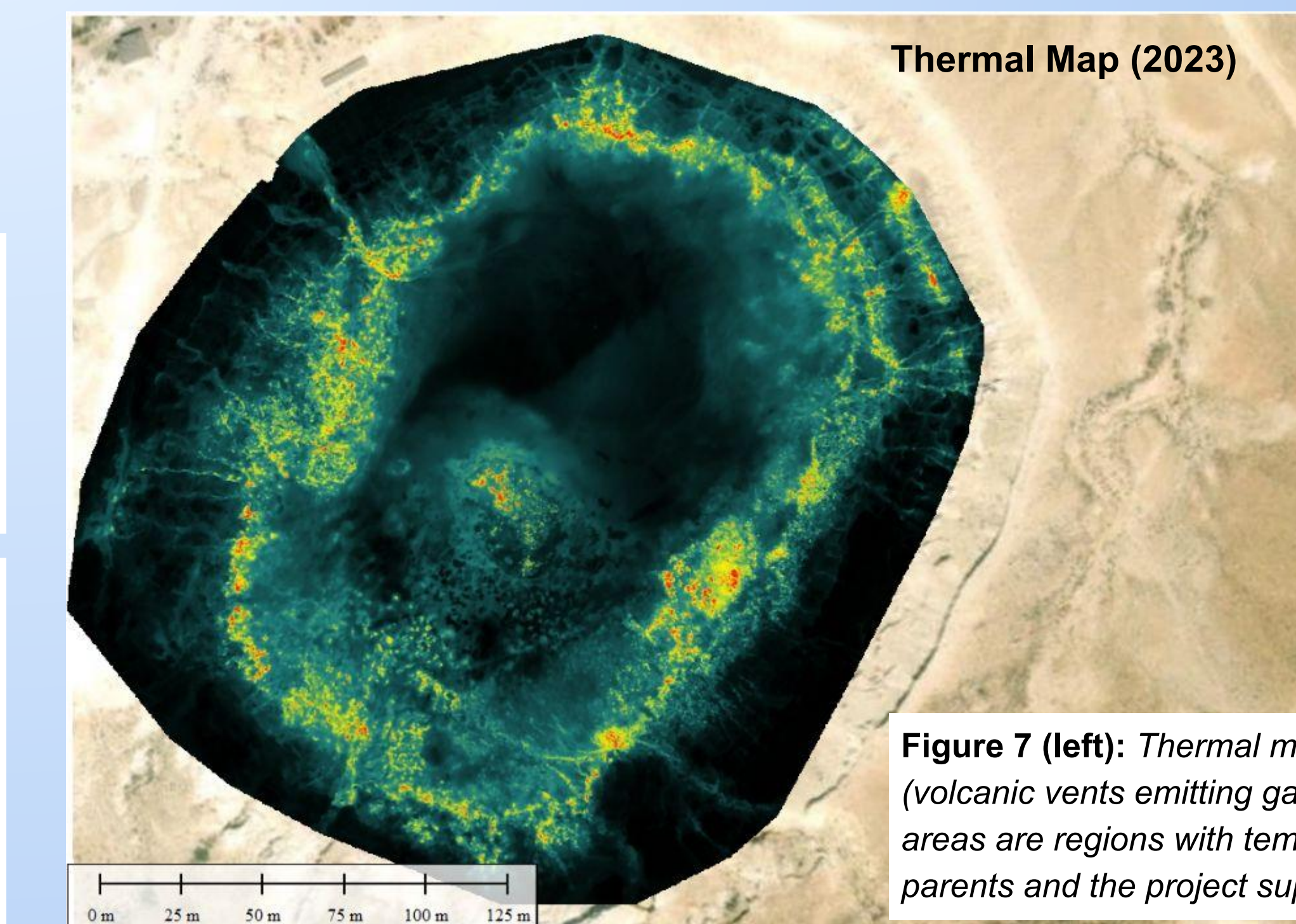


Figure 7 (left): Thermal map created by drone thermal scanning in July 2023. Red shaded regions are fumaroles (volcanic vents emitting gas and vapor). Yellow and green areas are cooler temperature regions, while black shaded areas are regions with temperatures equal to atmospheric. (Right): Hofstra students and high school student with her parents and the project supervisor Dr. Marsellos at Sousaki volcano.

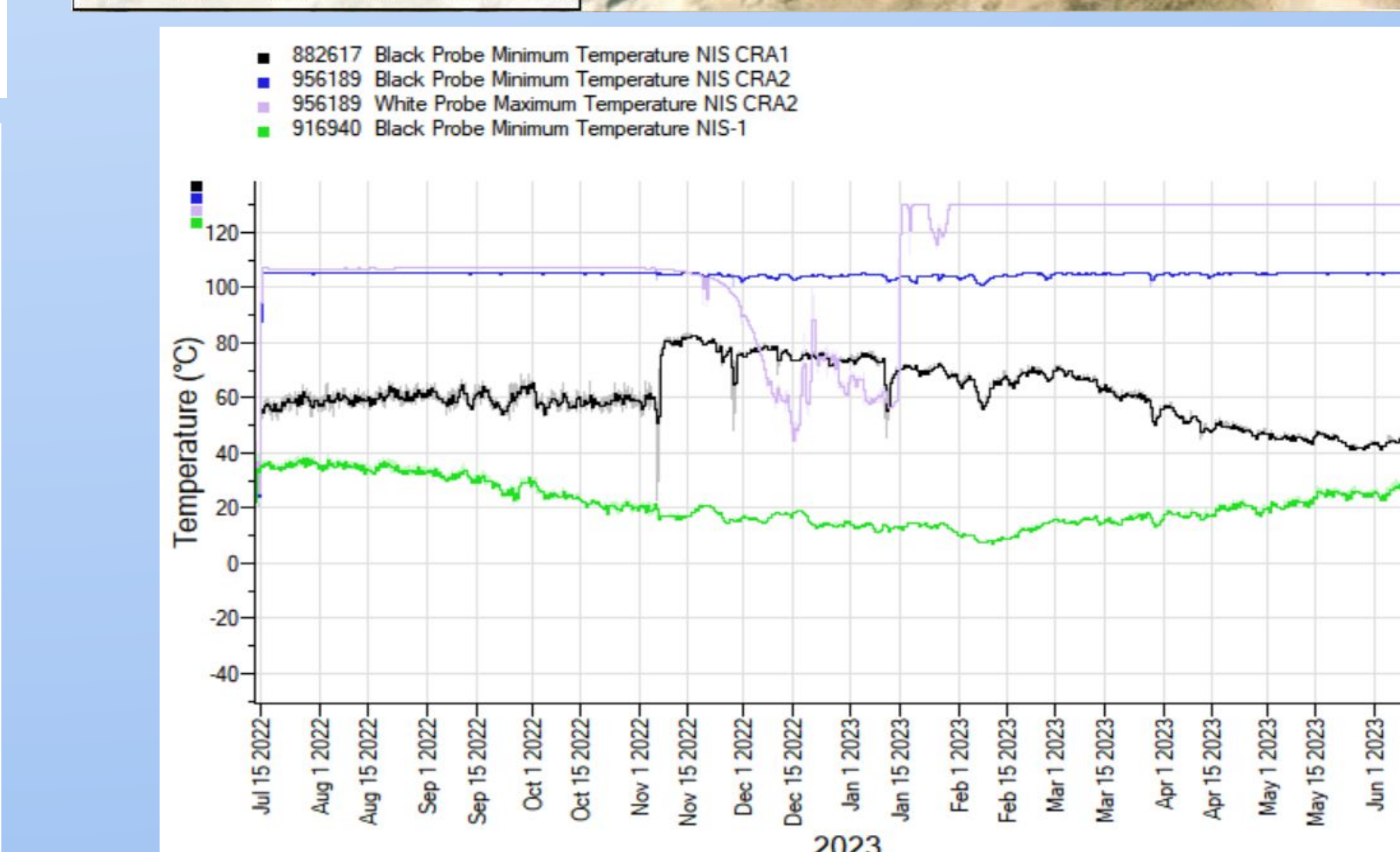


Figure 8: Hourly temperature logging from Nisyros Volcanic (Stefanos Crater). Temperature was recorded at the surface inside of the crater at an active fumarole (volcanic vent emitting gas and vapor; black, purple, and blue), and outside (green) for the time interval between the summers of 2022 and 2023 (956189 white probe shows malfunction).

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